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Engineering Applied Mathematics IIB for Three Players with Six Number of Jester Cards on Differential Calculus of Function of Several Variables–Level 5

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Abstract

This work presents 24 number of cards from applied mathematics IIB degree level mainly focused on differential calculus of function of several variables. The jester cards are A, B, C, D, E and F.

1 Introduction

At present, the decline in the reading community has reached a worrisome level. For many reasons, people have stopped sitting down and reading. For instance, wasting time on useless social media and in trivial places are among the many minor reasons. In 2025, Jenber proposed a solution to this through a major share company established under the title "Zero or One Trading Share Company." This share company believes that the problem mentioned above can be solved in the following ways. We found the solution included in this writing to be convincing (for more information, see the foundation document of 0 or 1 game [4]). The solution presented by this share company is briefly summarized below:

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(1) Making reading or studying entertaining, profitable, and rewarding. One of the methods for studying or acquiring new knowledge is understanding things by associating them with their equivalents. For example, an English word has many synonyms. Trying to memorize them by associating them is considered one of the best methods. However, to make this exercise unforgettable, doing it in the form of a game while having fun and discussing in groups makes it preferable.There are scientific studies conducted around studying English vocabulary by preparing cards (for example, see [1, 2, 3, 5, 6, 7, 8, 9, 10, 13]).

(2) Understanding things by associating them according to their types stimulates thinking, makes one inquisitive, sparks discussions with friends, and even drives one to search for reference books. And when done this way, it takes a much longer time to forget that thing.

(3) Furthermore, if this association is done enjoyably in the form of a game with friends, and if a bonus/reward is given to the winner to encourage knowledge and learning, the game will become popular.

(4) Developing various types of entertaining games to execute the solutions mentioned in 1-3 above.

In this paper, we have prepared cards for the 0 or 1 game. We have briefly described the number of cards as follows. Let the number of jester cards be 6 and the number of players be $k = 3$. Therefore, the total number of cards, $n = 8 \times 3 + 6 = 30$. Thus, including the jester cards, there are a total of 30 cards. The player cards will be $30 - 6 = 24$ cards; 24 is a multiple of 8 (for more information, see the 0 or 1 game foundation document [4]). The jester cards are A, B, C, D, E and F.

2 Preliminary

Many engineering systems depend on more than one independent variable. For example, temperature may depend on spatial coordinates, electric potential may vary with position, and mechanical stress may depend on several geometric and physical parameters. Differential calculus of functions of several variables provides mathematical tools for describing and analyzing such systems.

The following concepts introduce the main ideas of differential calculus of functions of several variables, with examples related to engineering applications. The discussion follows the multivariable calculus material in [11].

2.1 Functions of Several Variables and Their Domains

A function of two variables assigns a real number to each admissible ordered pair (x, y):

$$z = f(x, y).$$

In engineering, such a function may represent temperature, electrical power, mechanical stress, or manufacturing cost as a function of two parameters. The domain consists of all input pairs for which the function is defined and, where relevant, physically meaningful.

Example. Consider

$$T(x,y) = \sqrt{36 - x^2 - 4y^2}.$$

For T to be real-valued, the radicand must be nonnegative:

$$36 - x^2 - 4y^2 \geqslant 0.$$

Thus the domain is the elliptical region

$$x^2 + 4y^2 \leqslant 36.$$

The boundary is included because the square root of zero is defined. See [12, Section 4.1].

2.2 Limits and Continuity

The limit of $f(x,y)$ as (x,y) approaches (a,b) is L if $f(x,y)$ approaches L independently of the path used to approach (a,b). The function is continuous at (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

Checking one path alone does not prove that a multivariable limit exists. However, two paths giving different limiting values prove that the limit does not exist.

Example. Let

$$f(x,y) = \frac{xy}{x^2 + y^2}.$$

Along $y = x$, we have $f(x,x) = \dfrac{1}{2}$ for $x \neq 0$, whereas along $y = 0$, we have $f(x,0) = 0$. Since these values differ, the limit at the origin does not exist. See [12, Section 4.2].

2.3 Partial Derivatives and Their Engineering Interpretation

For $z = f(x,y)$, the first-order partial derivatives are

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}.$$

To calculate f_x, treat y as constant; to calculate f_y, treat x as constant. These derivatives measure the instantaneous rate of change of the output when one input changes while the other is held fixed.

Example. Electrical power in a simple model is $P(V,I) = VI$. Therefore,

$$\frac{\partial P}{\partial V} = I, \qquad \frac{\partial P}{\partial I} = V.$$

Higher-order partial derivatives include f_{xx}, f_{yy}, and mixed derivatives such as f_{xy}. If the relevant second partial derivatives are continuous in a neighborhood, then $f_{xy} = f_{yx}$. See [12, Section 4.3].

2.4 The Chain Rule for Dependent Variables

If $z = f(x, y)$, where $x = x(t)$ and $y = y(t)$, then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

The rule is useful when several quantities change simultaneously with time.

Example. Let

$$z = x^2 + 3y, \qquad x = t^2 + 1, \qquad y = 2t.$$

At $t = 1$, we have $x = 2$, $dx/dt = 2$, and $dy/dt = 2$. Hence

$$\frac{dz}{dt} = 2x\frac{dx}{dt} + 3\frac{dy}{dt} = 2(2)(2) + 3(2) = 14.$$

See [12, Section 4.5].

2.5 Tangent Planes and Linear Approximation

For a differentiable function $z = f(x, y)$, the tangent plane at $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Near (a, b), the function can be approximated by

$$f(a + \Delta x, b + \Delta y) \approx f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

Example. The diagonal of a rectangle is

$$d(x, y) = \sqrt{x^2 + y^2}.$$

At $(3, 4)$, $d = 5$ and

$$dd \approx \frac{3}{5}dx + \frac{4}{5}dy.$$

If $dx = 0.1$ and $dy = -0.2$, then

$$dd \approx \frac{3}{5}(0.1) + \frac{4}{5}(-0.2) = -0.10.$$

Thus the estimated new diagonal is 4.90. See [12, Section 4.4].

2.6 Implicit Differentiation

For a relation $F(x, y) = 0$, if $F_y \neq 0$, then locally

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Example. For

$$x^2 + xy + y^2 = 7,$$

implicit differentiation gives

$$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0,$$

so

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y}.$$

At $(1,2)$, the slope is $-4/5$. See [12, Section 4.5].

2.7 Gradient and Directional Derivatives

The gradient of $f(x,y)$ is

$$\nabla f(x,y) = \big(f_x(x,y), f_y(x,y)\big).$$

It points in the direction of greatest instantaneous increase of f. For a unit vector $\mathbf{u} = (u_1, u_2)$, the directional derivative is

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

Example. Let $f(x,y) = x^2 + y^2$. At $(1,2)$,

$$\nabla f(1,2) = (2,4), \qquad |\nabla f(1,2)| = 2\sqrt{5}.$$

The maximum directional derivative occurs in the direction $(1,2)/\sqrt{5}$. See [12, Section 4.6].

2.8 Total Differentials and Error Estimation

For a differentiable function $z = f(x,y)$, the total differential is

$$dz = f_x\,dx + f_y\,dy.$$

For small input errors, the differential estimates the resulting output error: $\Delta z \approx dz$.

Example. The volume of a cylinder is $V(r,h) = \pi r^2 h$, so

$$dV = 2\pi r h\,dr + \pi r^2\,dh.$$

This estimates how small errors in radius and height affect the calculated volume. See [12, Section 4.4].

2.9 Critical Points and Unconstrained Optimization

For a differentiable function of two variables, a critical point occurs where

$$f_x = 0, \qquad f_y = 0.$$

Define

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

At a critical point:

- If $D > 0$ and $f_{xx} > 0$, there is a local minimum.
- If $D > 0$ and $f_{xx} < 0$, there is a local maximum.
- If $D < 0$, there is a saddle point.
- If $D = 0$, the test is inconclusive.

Example. Consider

$$C(x, y) = x^2 + 2y^2 - 4x + 8y.$$

Solving $C_x = C_y = 0$ gives $(x, y) = (2, -2)$. Since $C_{xx} = 2$, $C_{yy} = 4$, and $C_{xy} = 0$, we have $D = 8 > 0$ and $C_{xx} > 0$. Thus the point is a minimum, with

$$C(2, -2) = -12.$$

See [12, Section 4.7].

2.10 Constrained Optimization and Lagrange Multipliers

To optimize $f(x, y)$ subject to $g(x, y) = c$, the Lagrange multiplier method uses

$$\nabla f = \lambda \nabla g, \qquad g(x, y) = c.$$

Example. A rectangular enclosure has one side along a wall, so fencing is required for only three sides. If x is the depth and y is the length along the wall, then $2x + y = 40$. The area is $A = xy$. Substituting $y = 40 - 2x$ gives

$$A(x) = 40x - 2x^2.$$

Since $A'(x) = 40 - 4x = 0$ gives $x = 10$, we obtain $y = 20$ and maximum area $A = 200$ square units. See [12, Section 4.8].

3 Jester and Player Cards

3.1 Instructions

Each question has six answer choices, labelled A, B, C, D, E, and F. Exactly one choice is correct for each question, and the other five are incorrect.

During each play, a student selects one answer choice. A correct selection earns a score of 1, while an incorrect selection earns a score of 0.

3.2 Jester Cards

The jester cards contain the texts A, B, C, D, E and F.

3.3 Player Cards

Question 1. Electrical power sensitivity. The power dissipated by a resistor is $P(V, I) = VI$. At $V = 24$ volts and $I = 3$ amperes, what is the rate of change of power with respect to current when voltage is constant?

(A) 3 W/A

(B) 8 W/A

(C) 12 W/A

(D) 24 W/A

(E) 27 W/A

(F) 72 W/A

Question 2. Feasible operating region. A sensor model is $T(x, y) = \sqrt{36 - x^2 - 4y^2}$. What is the maximum possible value of $|x|$ when $y = 2$?

(A) 2

(B) $2\sqrt{2}$

(C) $2\sqrt{3}$

(D) 4

(E) $2\sqrt{5}$

(F) 6

Question 3. Path-dependent limit. Let

$$F(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

What can be concluded about $\lim_{(x,y)\to(0,0)} F(x, y)$?

(A) The limit is 0.

(B) The limit is 1.

(C) The limit is $\frac{1}{2}$.

(D) The limit is -1.

(E) The limit is $+\infty$.

(F) The limit does not exist.

Question 4. Continuity of a signal model. Let

$$S(x, y) = \begin{cases} \dfrac{\sin(xy)}{xy}, & xy \neq 0, \\ 1, & xy = 0. \end{cases}$$

Which statement is correct?

(A) S is discontinuous at every point where $x = 0$.

(B) S is continuous everywhere in $\mathbb{R}^2$.

(C) S is continuous only when $x = y$.

(D) S is discontinuous at the origin but continuous elsewhere.

(E) S is undefined whenever $xy = 0$.

(F) S is continuous only when $xy \neq 0$.

Question 5. Sensitivity of a mechanical response. Let $R(x, y) = x^2y + y^3$. At $(2, -1)$, what is the rate of change of R with respect to x, holding y constant?

(A) -8

(B) -6

(C) -4

(D) -2

(E) 2

(F) 4

Question 6. Mixed partial derivatives. Let $T(x, y) = e^{xy}$. What is $T_{xy}(0, 1)$?

(A) 0

(B) e

(C) 2

(D) 1

(E) $1 + e$

(F) -1

Question 7. Rate of change in a moving system. Let

$$z = x^2 + 3y, \qquad x = t^2 + 1, \qquad y = 2t.$$

What is dz/dt at $t = 1$?

(A) 6

(B) 8

(C) 10

(D) 12

(E) 14

(F) 16

Question 8. Tangent-plane approximation. Let $z = \ln(x + 2y)$. Which equation is the tangent plane at $(1, 1, \ln 3)$?

(A) $z = \ln 3 + \dfrac{x-1}{3} + \dfrac{2(y-1)}{3}$

(B) $z = \ln 3 + \dfrac{2(x-1)}{3} + \dfrac{y-1}{3}$

(C) $z = \ln 3 + (x-1) + 2(y-1)$

(D) $z = \ln 3 - \dfrac{x-1}{3} - \dfrac{2(y-1)}{3}$

(E) $z = \ln 3 + \dfrac{x+y-2}{3}$

(F) $z = \ln 3 + \dfrac{x-1}{2} + \dfrac{y-1}{2}$

Question 9. Estimating a measured diagonal. Let $d(x,y) = \sqrt{x^2+y^2}$. Initially, $x = 3$ cm and $y = 4$ cm. If x increases by 0.1 cm and y decreases by 0.2 cm, what is the linear approximation of the new diagonal?

(A) 4.80 cm

(B) 4.90 cm

(C) 5.00 cm

(D) 5.02 cm

(E) 5.10 cm

(F) 5.20 cm

Question 10. Implicit differentiation of a constraint. The curve $x^2 + xy + y^2 = 7$ passes through $(1,2)$. What is the slope dy/dx there?

(A) $-\dfrac{5}{4}$

(B) -1

(C) $-\dfrac{4}{5}$

(D) $-\dfrac{2}{5}$

(E) $\dfrac{4}{5}$

(F) $\dfrac{5}{4}$

Question 11. Maximum directional rate of change. Let $f(x,y) = x^2y$. What is the maximum directional derivative at $(1,2)$?

(A) 1

(B) 2

(C) 4

(D) $4\sqrt{2}$

(E) 5

(F) $\sqrt{17}$

Question 12. Directional derivative of a potential field. Let $\phi(x, y) = \ln(x^2 + y^2)$. At $(1, 1)$, determine the directional derivative in the direction from the origin toward $(1, 1)$.

(A) 0

(B) 1

(C) $\sqrt{2}$

(D) 2

(E) $2\sqrt{2}$

(F) $-\sqrt{2}$

Question 13. Tangent to a contour curve. A contour curve is defined by $x^2 + 3y^2 = 4$. What is the slope of the tangent line at $(1, 1)$?

(A) -3

(B) -2

(C) $-\dfrac{1}{3}$

(D) $\dfrac{1}{3}$

(E) 1

(F) 3

Question 14. Temperature change along a trajectory. Let $T(x, y) = 80 - x^2 - 2y^2$. A particle passes through $(2, 1)$ with velocity vector $\mathbf{v} = (3, -1)$. What is the instantaneous rate of change of temperature experienced by the particle?

(A) -16

(B) -12

(C) -8

(D) -4

(E) 4

(F) 8

Question 15. Error propagation in a cylinder. The volume of a cylinder is $V = \pi r^2 h$. At $r = 2$ cm and $h = 10$ cm, the radius increases by 0.01 cm, while the height decreases by 0.05 cm. Using the total differential, estimate the change in volume.

(A) -0.2π cm^3

(B) 0 cm^3

(C) 0.1π cm^3

(D) 0.2π cm^3

(E) 0.4π cm^3

(F) 0.6π cm^3

Question 16. Error in a calculated measurement. Let $Q(x, y) = xy^2$. At $x = 2$ and $y = 3$, suppose $dx = 0.01$ and $dy = -0.02$. What is the approximate change dQ?

(A) -0.33

(B) -0.24

(C) -0.15

(D) -0.09

(E) 0.15

(F) 0.24

Question 17. Rate of decay of concentration. Let $C(x, t) = 10e^{-0.2t} \cos x$. What is the rate of change of concentration with respect to time at $x = \pi/3$ and $t = 0$?

(A) -2

(B) -1

(C) -0.5

(D) 0

(E) 0.5

(F) 1

Question 18. Minimizing a cost function. Let $C(x, y) = x^2 + 2y^2 - 4x + 8y$. For unrestricted real x and y, what is the minimum value of C?

(A) -16

(B) -14

(C) -12

(D) -10

(E) -8

(F) 0

Question 19. Classifying a critical point. Let $U(x, y) = x^2 - 4y^2$. What type of critical point occurs at the origin?

(A) Strict local minimum

(B) Strict local maximum

(C) Saddle point

(D) Neither a critical point nor a stationary point

(E) A flat region of global minima

(F) A point where the function is undefined

Question 20. Optimizing a rectangular enclosure. A rectangular enclosure is built beside a straight wall, so fencing is needed for only three sides. There are 40 metres of fencing available. What is the maximum possible enclosed area?

(A) $100\ \text{m}^2$

(B) $150\ \text{m}^2$

(C) $180\ \text{m}^2$

(D) $196\ \text{m}^2$

(E) $200\ \text{m}^2$

(F) $400\ \text{m}^2$

Question 21. Constrained production optimization. Let $P(x, y) = x^2 y$, where $x > 0$, $y > 0$, and $x + y = 6$. What is the maximum value of P?

(A) 24

(B) 30

(C) 32

(D) 36

(E) 48

(F) 64

Question 22. Finding a minimum operating cost. Let

$$C(x, y) = x^2 + xy + y^2 - 6x - 6y.$$

For unrestricted real x and y, at which point does the cost attain its minimum?

(A) $(1, 1)$

(B) $(2, 2)$

(C) $(3, 3)$

(D) $(0, 3)$

(E) $(3, 0)$

(F) $(2, -2)$

Question 23. Continuity of a vibration model. Let

$$g(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\dfrac{1}{\sqrt{x^2 + y^2}}\right), & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

What is $\lim\limits_{(x,y)\to(0,0)} g(x, y)$?

(A) -1

(B) 0

(C) 1

(D) The limit does not exist.

(E) $+\infty$

(F) $-\infty$

Question 24. Tangent plane to a spherical surface. A sphere is described by $x^2 + y^2 + z^2 = 14$. Which equation is the tangent plane at $(1, 2, 3)$?

(A) $x + 2y + 3z = 7$

(B) $2x + 4y + 6z = 7$

(C) $x + y + z = 6$

(D) $x + 2y + 3z = 14$

(E) $x + 2y + 3z = 6$

(F) $2x + y + 3z = 14$

Scoring rule:

$$\text{Score} = \begin{cases} 1, & \text{if the selected choice is correct,} \\ 0, & \text{if the selected choice is incorrect.} \end{cases}$$

3.4 Answer Key

Question	Answer	Main concept
1	D	Partial derivatives
2	E	Domain of a function
3	F	Multivariable limits
4	B	Continuity
5	C	Partial derivatives
6	D	Mixed partial derivatives
7	E	Chain rule
8	A	Tangent plane
9	B	Linear approximation
10	C	Implicit differentiation
11	F	Gradient magnitude
12	C	Directional derivative
13	C	Tangent to a level curve
14	C	Chain rule along a trajectory
15	D	Total differential
16	C	Error estimation
17	B	Partial derivative with respect to time
18	C	Unconstrained minimization
19	C	Saddle point
20	E	Constrained optimization
21	C	Constrained optimization
22	B	Minimum cost
23	B	Limit and continuity
24	D	Tangent plane to an implicit surface

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